MOHIT SINGH RAJPUT

ML & Generative AI Engineer

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TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript

ML / AI: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, LLMs, RAG, Prompt Engineering

Frameworks, Libraries & Tools: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, LangChain, LangGraph, NumPy, Pandas, Matplotlib, Seaborn, FastAPI, HTML, CSS, Amazon EC2, Git, GitHub, SQL

SUMMARY

ML & Generative AI Engineer with end-to-end experience across NLP, computer vision, and applied deep learning, seeking to apply expertise in LLM applications, RAG pipelines, and multi-agent systems to build tested, documented, and production-ready AI solutions.

EXPERIENCE

AI/ML Intern — Labmentix *June – July 2026*

Multi-Agent LLM Systems & Computer Vision | RAG, FastAPI, TensorFlow, TFLite

- Completed a 2-month internship (Jun 1 – Jul 31, 2026) focused on applied ML and deep learning projects.
- Built and evaluated two deep learning architectures (a from-scratch CNN and a fine-tuned MobileNetV2) for facial emotion recognition, covering data preprocessing, augmentation, training, and TFLite quantization for on-device deployment.
- Designed a multi-agent LLM customer support system with intent-based routing, RAG retrieval, and a FastAPI backend, deployed live across Render and Vercel.
- Applied consistent evaluation discipline across both projects, including held-out test sets, quantitative benchmarking of every model variant, and automated tests.

Machine Learning Intern — CodeAlpha *June 2026*

Project-Based ML Internship | Python, Scikit-learn, Pandas

- Completed a project-based machine learning internship covering end-to-end data preprocessing, feature engineering, and model evaluation.
- Applied supervised and unsupervised techniques (classification, clustering) to real-world datasets in Python with scikit-learn and pandas.

PROJECTS

Multi-Agent LLM Customer Support Platform | *Python, FastAPI, Next.js, FAISS, Sentence-Transformers, Groq/OpenAI, SQLAlchemy, JWT, Twilio, SendGrid* Live Demo

- Architected a 5-agent customer support system (billing, technical, product, complaints, FAQ) behind a two-stage intent router, reaching 90% intent-classification and agent-routing accuracy on a multilingual evaluation set.
- Built a RAG pipeline (FAISS, Sentence-Transformers) over an 8-document knowledge base, and a FastAPI backend with REST endpoints for auth, chat, analytics, and admin.
- Integrated Twilio (WhatsApp) and SendGrid (email) for automated ticket escalation, and deployed the full stack to production on Render and Vercel with retry logic for LLM call resilience.

DeepFER — Facial Emotion Recognition | *Python, TensorFlow/Keras, OpenCV, TFLite, Flask, pytest* Live Demo

- Trained and benchmarked two CNN architectures for 7-class facial emotion recognition on FER-2013 (35,887 images), reaching 52.27% and 59.46% accuracy on a held-out test set.
- Quantized both models to TFLite (float32, int8, full int8); full-integer quantization cut latency by up to 108x with no accuracy loss.
- Shipped a Flask web app for static-image and live browser-camera inference, backed by an automated pytest suite and reproducible evaluation artifacts.

Shopper Spectrum — Customer Segmentation & Recommendation Engine | *Python, Scikit-learn, Pandas, Streamlit, Joblib* Live Demo

- Engineered an RFM feature pipeline from 541,909 UK retail transactions across 4,338 customers with outlier treatment and normalization.
- Benchmarked four clustering algorithms and selected K-Means (k=4) to segment customers into High-Value, Regular, Occasional, and At-Risk groups.
- Built an item-based collaborative-filtering recommender and shipped both modules as an interactive Streamlit application.

Flipkart CSAT Prediction — Explainable ML | *Python, Scikit-learn, XGBoost, NLTK, SHAP* Live Demo

- Built a binary CSAT classification pipeline on 85,907 Flipkart support tickets, combining structured and TF-IDF text features, and tuned three classifiers via GridSearchCV.
- The final XGBoost model reached 72.6% accuracy and 0.78 ROC-AUC; applied SHAP for explainability and serialized the pipeline via Joblib for deployment.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science & Engineering *Aug 2023 – Mar 2027 (Expected)*

Shankara Institute of Technology — Jaipur, Rajasthan

CERTIFICATES / EXTRACURRICULAR

- Oracle Certified Foundations Associate — Agentic AI (Oracle University, Aug 2026)
- Deloitte Australia Data Analytics Job Simulation — Forage (Aug 8, 2026)
  - Completed a Deloitte job simulation involving data analysis and forensic technology
  - Created a data dashboard using Tableau
  - Used Excel to classify data and draw business conclusions
- Tata Data Visualisation: Empowering Business with Effective Insights — Forage (Aug 8, 2026)
  - Completed a simulation involving creating data visualizations for Tata Consultancy Services
  - Prepared questions for a meeting with client senior leadership
  - Created visuals for data analysis to help executives with effective decision making
- Walmart USA Advanced Software Engineering Virtual Experience Program — Forage (Aug 9, 2026)
  - Completed the Advanced Software Engineering Job Simulation, solving technical projects for a variety of teams at Walmart
  - Developed a novel version of a heap data structure in Java for Walmart's shipping department, showcasing strong problem-solving and algorithmic skills
  - Designed a UML class diagram for a data processor, considering different operating modes and database connections
  - Created an entity relationship diagram to design a new database accounting for all requirements provided by Walmart's pet department